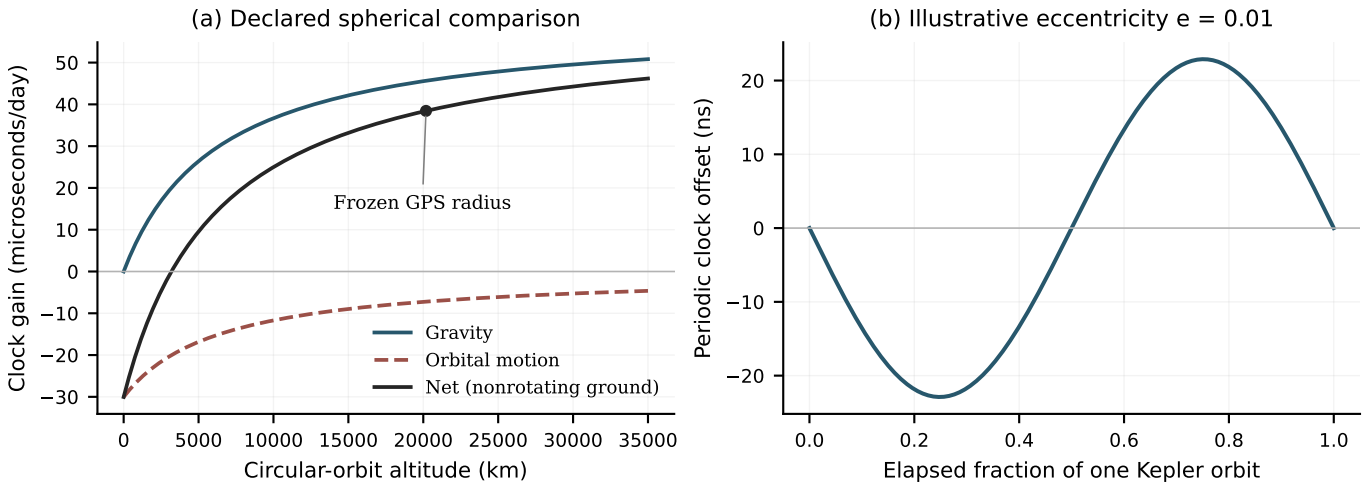
(a) Declared spherical comparison
Clock gain (microseconds/day)
50
40
30
20
10
0
−10
−20
−30
Frozen GPS radius
Gravity
Orbital motion
Net (nonrotating ground)
0
5000
10000
15000
20000
25000
30000
35000
Circular-orbit altitude (km)
(b) Illustrative eccentricity e = 0.01
Periodic clock offset (ns)
20
10
0
−10
−20
0.0
0.2
0.4
0.6
0.8
1.0
Elapsed fraction of one Kepler orbit